

THUMRAIT, Oman

WMO: 413140

Lat: 17.666N

Lon: 54.025E

Elev: 479

StdP: 95.71

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.2	11.9	-11.0	1.6	20.8	-8.2	2.0	22.3	11.5	23.5	10.2	23.0	2.9	170	0.536

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	13.0	42.2	19.0	41.1	19.0	40.0	19.2	24.6	32.8	24.2	32.4	23.8	31.9	4.9	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.7	18.5	28.0	22.1	17.8	27.6	21.7	17.4	27.5	77.8	32.7	75.9	32.2	74.3	32.2	30.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.4	13.2	12.1	DB	7.6	44.2	1.3	0.9	6.6	44.9	5.8	45.4	5.1	46.0	4.1	46.6	(4)
(5)			WB	1.5	25.8	1.4	1.6	0.5	27.0	-0.4	27.9	-1.2	28.8	-2.2	29.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.9	19.3	21.6	25.2	28.9	31.7	32.9	30.9	30.3	30.5	27.6	23.6	20.2	(6)	
	DBStd	5.05	2.56	2.77	2.34	1.88	1.74	2.04	2.45	2.22	2.26	1.81	1.72	2.28	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	36	21	6	0	0	0	0	0	0	0	0	0	9	(9)	
	CDD10.0	6173	287	325	471	568	673	686	649	628	614	546	409	317	(10)	
	CDD18.3	3167	49	98	212	318	414	436	391	370	364	288	159	68	(11)	
	CDH23.3	38462	389	891	2182	3945	5788	6170	4788	4282	4568	3391	1533	534	(12)	
(13)	CDH26.7	21342	63	287	1017	2257	3561	3893	2666	2344	2756	1869	542	88	(13)	
(14)	Wind	WSAvg	5.8	3.7	4.6	5.6	5.7	6.2	7.4	9.6	9.2	6.0	4.3	3.4	3.3	(14)
(15) Precipitation	PrecAvg	23	0	3	3	5	3	2	0	2	0	1	0	0	(15)	
	PrecMax	145	5	58	40	116	74	26	1	39	8	22	9	7	(16)	
	PrecMin	0	0	0	0	0	0	0	0	0	0	0	0	0	(17)	
	PrecStd	38	1	10	9	21	13	5	0	7	1	4	2	1	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.2	33.4	36.6	39.2	42.3	43.8	43.1	42.4	41.9	38.0	33.1	30.3	(19)	
		MCWB	15.9	16.3	17.1	17.8	18.7	19.1	18.8	20.1	19.1	17.5	16.3	16.7	(20)	
	2%	DB	28.2	31.4	34.8	38.0	41.0	42.5	41.7	40.5	40.5	36.7	31.9	28.7	(21)	
		MCWB	16.1	16.0	16.6	18.0	18.4	18.8	18.9	20.6	19.0	17.2	16.6	16.8	(22)	
	5%	DB	27.0	29.9	33.2	36.8	39.8	41.2	39.5	38.3	39.2	35.5	30.9	27.5	(23)	
		MCWB	16.1	16.1	16.3	17.7	18.5	19.0	20.1	21.6	18.9	17.0	16.7	16.6	(24)	
	10%	DB	25.7	28.4	31.8	35.6	38.5	39.8	36.8	36.0	37.6	34.2	29.8	26.4	(25)	
		MCWB	15.6	15.9	16.4	17.7	18.7	19.3	22.4	22.4	19.1	16.9	16.8	16.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.4	20.7	21.5	23.4	24.4	25.4	25.2	24.6	24.0	23.1	21.7	20.5	(27)	
		MCDB	26.0	27.4	29.0	30.1	32.0	33.3	33.8	34.3	32.6	28.6	26.2	25.3	(28)	
	2%	WB	18.4	19.4	20.7	22.4	23.7	24.7	24.5	24.0	23.1	22.1	21.0	19.2	(29)	
		MCDB	24.7	25.9	27.7	28.8	31.0	32.3	33.0	33.3	31.7	27.8	25.8	24.4	(30)	
	5%	WB	17.7	18.3	20.0	21.7	23.2	24.3	24.0	23.6	22.5	21.2	20.0	18.4	(31)	
		MCDB	23.7	25.1	26.7	28.5	30.2	32.0	32.6	32.6	30.6	27.2	25.2	24.2	(32)	
	10%	WB	17.0	17.5	19.4	21.1	22.7	23.9	23.6	23.2	22.0	20.4	19.1	17.6	(33)	
		MCDB	22.9	24.6	26.3	28.5	29.7	31.6	31.6	31.9	29.5	26.8	25.3	23.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	12.4	13.2	13.0	13.2	13.1	13.0	11.1	11.5	13.5	13.3	12.4	12.0	(35)	
		MCDBR	13.3	14.7	14.7	14.4	15.0	15.2	14.9	14.9	15.1	14.3	13.3	12.9	(36)	
		MCWBR	4.8	5.4	5.6	5.6	5.6	5.5	4.4	3.5	5.0	5.4	5.0	5.0	(37)	
	5% WB	MCDBR	11.7	12.0	10.8	12.0	12.7	12.4	10.8	11.2	12.8	11.8	11.3	11.6	(38)	
		MCWBR	4.4	4.6	3.7	3.9	3.9	3.4	2.5	2.8	3.7	4.4	4.5	4.8	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.382	0.404	0.452	0.486	0.560	0.707	0.754	0.684	0.546	0.414	0.397	0.384	(40)	
	taud		2.147	2.071	1.933	1.876	1.715	1.461	1.427	1.547	1.771	2.087	2.133	2.144	(41)	
	Ebn at Noon		895	895	861	831	762	651	622	672	770	872	872	879	(42)	
	Edh at Noon		144	163	194	207	240	306	318	284	225	159	145	141	(43)	
(44) All-Sky Solar Radiation	RadAvg		5.37	6.15	6.84	7.26	7.37	6.88	6.19	6.36	6.66	6.36	5.51	5.11	(44)	
	RadStd		0.14	0.13	0.23	0.19	0.18	0.27	0.27	0.15	0.13	0.15	0.16	0.15	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	-0.23	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	+0.15	N/A	-1.52	N/A	N/A	-0.30	N/A	N/A	N/A	N/A

Nomenclature: See separate page